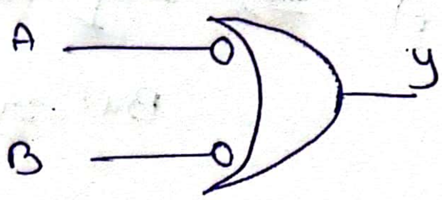
Alternate logic gates (or) Alternate symbols of gates (or) Bubbled gates :-

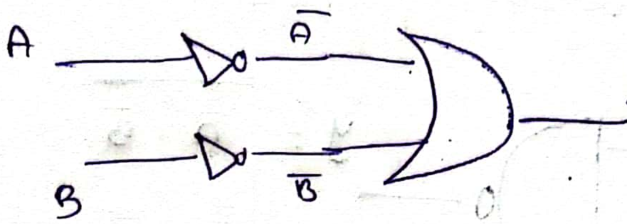
Bubbled OR gate :-

Bubble indicates "NOT" operation.

OR gate $(y) = A + B$.



$\Rightarrow y = \bar{A} + \bar{B}$



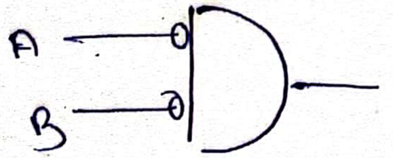
$y = \bar{A} + \bar{B}$

(or)

$y = \overline{AB}$ NAND

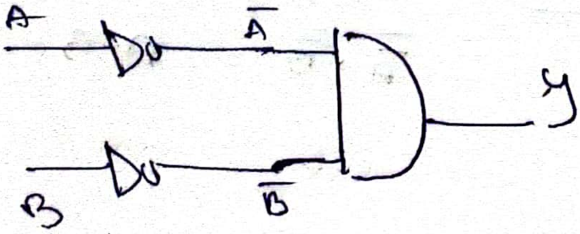
$= \bar{A} + \bar{B}$

Bubbled AND gate :-



$y = \bar{A} \cdot \bar{B}$

(or)



$y = \bar{A} \cdot \bar{B}$

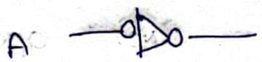
(or)

$= \overline{A + B} \Rightarrow$ NOR

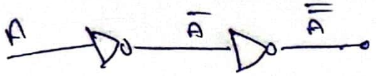
$\Rightarrow \bar{A} \cdot \bar{B}$

Bubbled NOT gate

$y = \bar{A}$



$\bar{\bar{A}} = A$

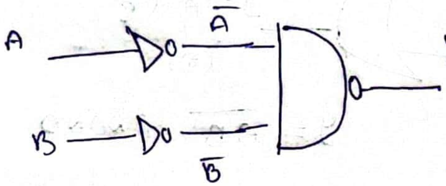


$y = A \Rightarrow$

$y = A$

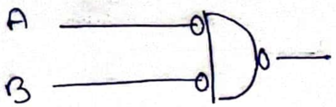
Buffer

Bubbled NAND gate :-



$y = \overline{\bar{A} \cdot \bar{B}}$

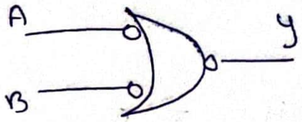
$= \bar{\bar{A}} + \bar{\bar{B}} = A + B$



$y = \overline{\bar{A} \cdot \bar{B}} = A + B$

OR gate

Bubbled NOR gate :-

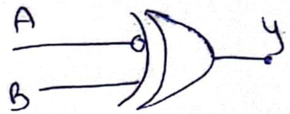


$y = \overline{\bar{A} + \bar{B}}$

$= \bar{\bar{A}} \cdot \bar{\bar{B}} = A \cdot B \Rightarrow$ AND

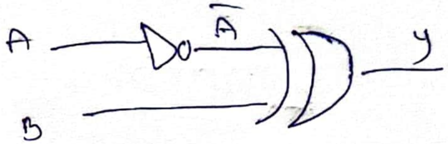
Bubbled X-OR gate :-

$y = A\bar{B} + \bar{A}B$



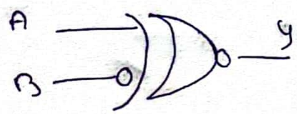
$y =$ ~~AB~~ $= \bar{A} \cdot \bar{B} + \bar{\bar{A}}B$

$= \bar{A} \cdot \bar{B} + AB$



$= \underbrace{\bar{A} \cdot \bar{B} + AB}_{\text{X-NOR gate.}}$

Bubbled X-NOR gate :-



$y = ?$

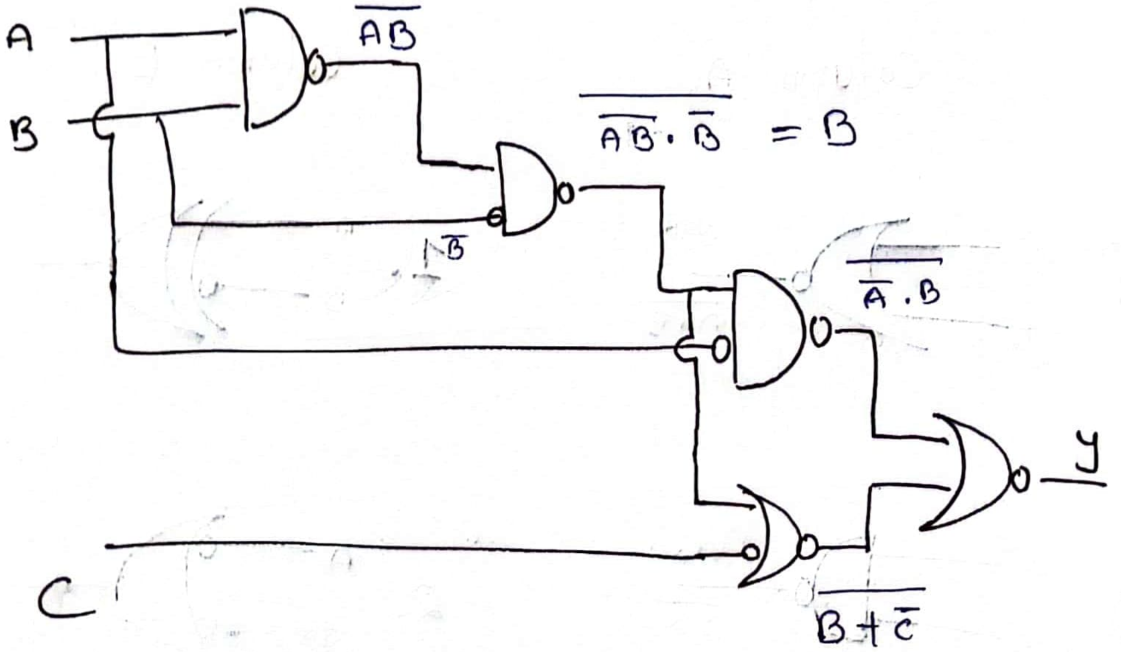
X-NOR

$AB + \bar{A} \cdot \bar{B}$

$y = A\bar{B} + \bar{A} \cdot \bar{\bar{B}}$

$= A\bar{B} + \bar{A}B \Rightarrow$ X-OR gate.

The simplified Boolean expression for the o/p y is ?



$$\overline{\overline{AB}\cdot\overline{B}} \Rightarrow \overline{\overline{AB}} + \overline{\overline{B}} = AB + B \Rightarrow B(1+A) \Rightarrow B$$

$$y = \overline{\overline{\overline{A}\cdot B} + \overline{B+\overline{C}}}$$

$$= \overline{\overline{\overline{A}\cdot B}} \cdot \overline{\overline{B+\overline{C}}} \Rightarrow (\overline{A}\cdot B)(B+\overline{C})$$

$$= \overline{A}\,\underbrace{B\cdot B}_{B} + \overline{A}\cdot B\cdot\overline{C} \Rightarrow \overline{A}\cdot B + \overline{A}\cdot B\cdot\overline{C}$$

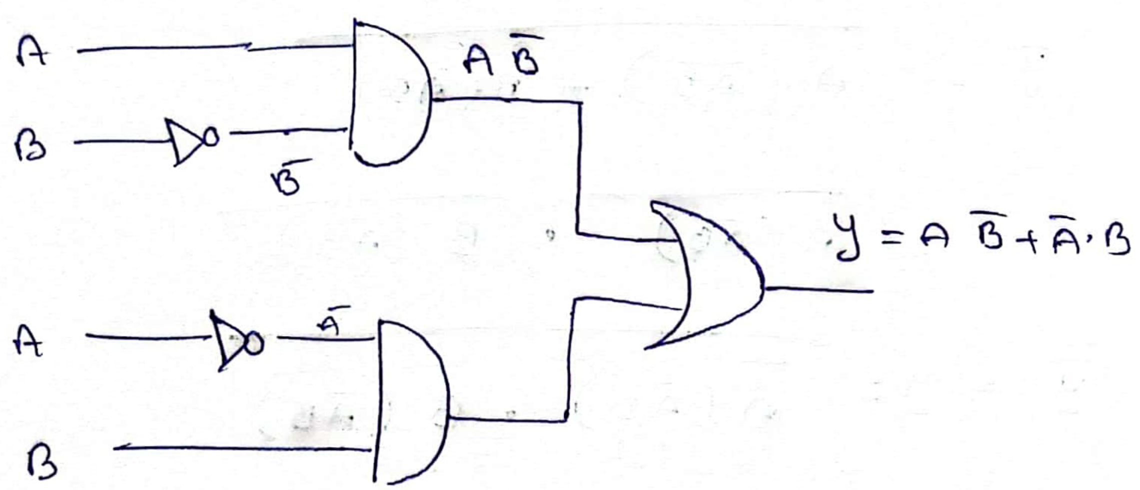
$$= B\left[\overline{A} + \overline{A}\cdot\overline{C}\right] \Rightarrow B\cdot\overline{A}\left[1+\overline{C}\right]$$

$$y = B\cdot\overline{A}$$

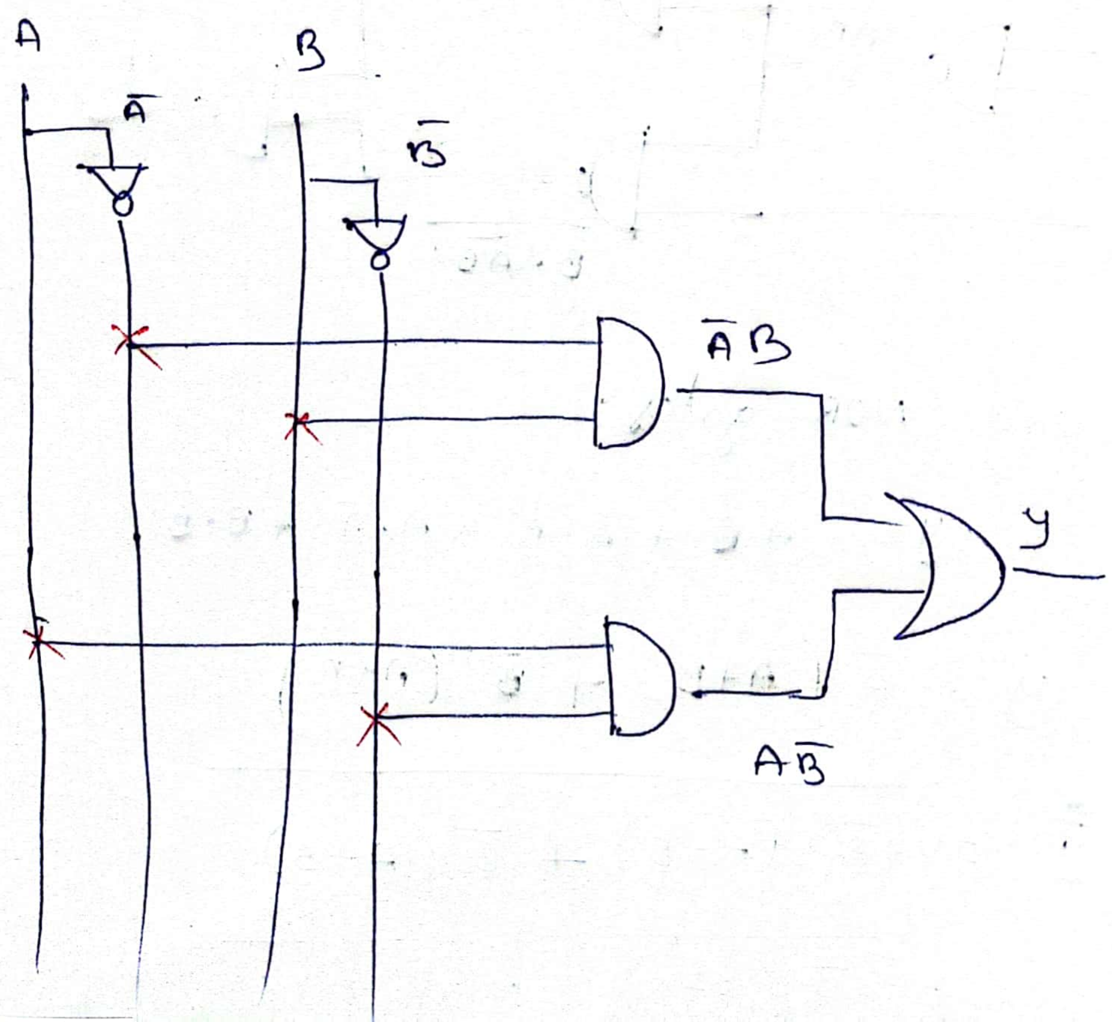
Implements of Boolean fn using logic gates :-

2) $Y = A\bar{B} + \bar{A}B$

using basic gates



(or)



by using NAND gates :-

$y = A\bar{B} + \bar{A}.B + A.\bar{A} + B.\bar{B}$

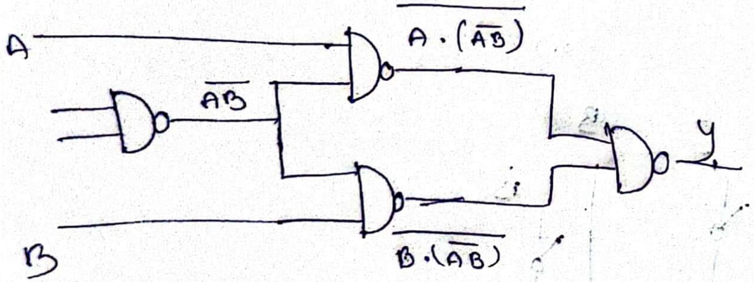
$= A(\bar{A}+\bar{B}) + B(\bar{A}+\bar{B})$

$y = A(\overline{AB}) + B(\overline{AB})$

$\bar{y} = \overline{A.(\overline{AB}) + B(\overline{AB})}$

$= \overline{A(\overline{AB})} \cdot \overline{B(\overline{AB})}$

$\bar{\bar{y}} = y = \overline{\overline{A(\overline{AB})} \cdot \overline{B(\overline{AB})}}$



using NOR gates :-

$y = A\bar{B} + \bar{A}\cdot B + A\cdot\bar{A} + B\cdot\bar{B}$

$y = \bar{A}.(A+B) + \bar{B}(A+B)$

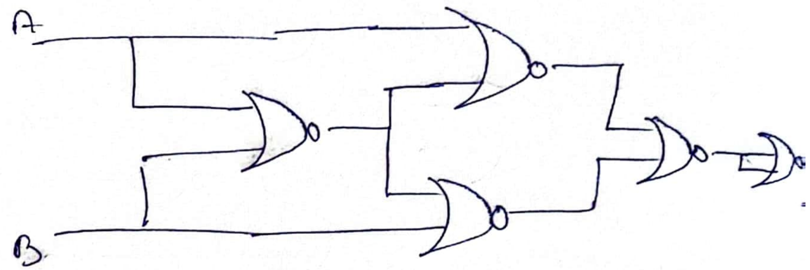
$\bar{y} = \overline{\bar{A}\cdot(A+B) + \bar{B}(A+B)}$

$= \overline{\bar{A}(A+B)} \cdot \overline{\bar{B}(A+B)}$

$= A + (\overline{A+B}) \cdot B + (\overline{A+B})$

$\bar{\bar{y}} = y = \overline{[A + (\overline{A+B})][B + (\overline{A+B})]}$

$= \overline{A + (\overline{A+B})} + \overline{(B + (\overline{A+B}))}$



5 NOR gates